



Vardan Pal

Software Engineering Intern | Full Stack Developer | Backend Engineer
B.Tech – Information Technology
2022 – 2026
National Institute of Technology, Srinagar

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PROFESSIONAL SUMMARY

Software Engineer with professional internship experience developing cloud-native backend systems and production-ready full-stack applications. Engineered 20+ RESTful API endpoints across healthcare, travel, and B2B platforms, integrating AI-powered OCR pipelines achieving ~98% extraction accuracy using OpenAI GPT-4o and Google Cloud Vision APIs. Proficient in MERN Stack, Next.js, Node.js, Express.js, TypeScript, Docker, MongoDB, PostgreSQL, secure API development, JWT-based authentication & authorization, scalable system architecture, database optimization, and Agile software development. Passionate about building secure, scalable, and high-performance software solutions for real-world applications.

EDUCATION

- National Institute of Technology, Srinagar** 2022 – 2026
Bachelor of Technology – Information Technology
– Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, System Design
- Army Public School** 2022
Class XII – CBSE | 87.4%
- Army Public School** 2020
Class X – CBSE | 91.4%

EXPERIENCE

- Noventiq (Umbrella Infotech)** Dec 2024 – Feb 2025
Software Engineering Intern – Cloud Native Applications Noida, India
– Engineered and delivered 8+ modular RESTful API endpoints for enterprise cloud-native applications using Node.js and Express.js; optimized middleware execution pipelines to reduce response latency by eliminating redundant database query patterns.
– Containerized 3 backend microservices using Docker, standardizing development and staging deployments while eliminating environment-specific runtime inconsistencies and improving deployment reliability.
– Implemented secure backend engineering practices including input validation, parameterized database queries, JWT-based authentication, and role-based authorization across 5+ service modules, strengthening overall application security.
– Collaborated within Agile/Scrum development cycles by conducting peer code reviews, refactoring legacy backend modules, resolving production-critical defects, and contributing to successful sprint deliveries.
– Optimized MongoDB database performance through schema restructuring, compound indexing strategies, and query optimization techniques, improving data retrieval efficiency for high-frequency REST API operations.

PROJECTS

- HealthEase – AI-Powered Telemedicine Platform | Final Year Project** Jan 2025 – Jun 2025
React.js, Node.js, Express.js, MongoDB, Docker, GPT-4o, Google Cloud Vision API, JWT, RBAC HealthEase GitHub
– Architected a multi-role telemedicine platform with service-oriented backend architecture by implementing separate RESTful API layers for Patient, Doctor, and Admin modules secured using JWT authentication and RBAC middleware.
– Designed and integrated an OCR pipeline using Google Cloud Vision API to digitize handwritten medical prescriptions, achieving ~98% extraction accuracy across structured and semi-structured prescription formats.
– Engineered an AI-powered healthcare assistant by integrating OpenAI GPT-4o with custom prompt engineering, enabling context-aware patient interactions while applying domain-specific guardrails for safe AI responses.
– Developed and secured 15+ RESTful API endpoints supporting appointments, prescriptions, medication reminders, and user management using JWT authentication, RBAC authorization, middleware validation, and secure API design.
– Containerized the complete application stack using Docker with environment-specific configurations, enabling reproducible deployments and production-ready application delivery.
- Mohishree Facility Services – Production B2B Industrial Marketplace** Jun 2024 – Sep 2024
Next.js, TypeScript, Node.js, PostgreSQL, Tailwind CSS, REST APIs Mohishree GitHub
– Architected and developed a production-ready B2B marketplace currently used by a live industrial services business; designed a normalized PostgreSQL schema with role-based access control for Client, Vendor, and Admin workflows.

- Developed a fully type-safe RESTful API layer in TypeScript with strict request–response contracts, enabling reliable integration between the Next.js SSR frontend and Node.js backend while eliminating runtime type inconsistencies.
- Implemented Server-Side Rendering (SSR) using Next.js with dynamic routing and static generation strategies to improve SEO, reduce time-to-interactive, and optimize first-page load performance.
- Engineered multi-step vendor onboarding workflows with document upload handling, Multer integration, form validation, and administrator approval pipelines supporting end-to-end business operations.
- Deployed and maintained the application for real-world business operations, supporting service discovery, vendor onboarding, quotation management, and administrative workflows.

•Wanderlust – Full-Stack Travel Listing Platform

Aug 2023 – Dec 2023

Node.js, Express.js, MongoDB, Passport.js, EJS, Cloudinary, Mapbox API

 GitHub

- Designed and implemented a production-grade MVC backend architecture with modular route handlers, middleware layers, and Mongoose data models, enabling scalable feature development.
- Engineered secure session-based authentication and authorization using Passport.js, bcrypt password hashing, persistent login sessions, and protected middleware across all application routes.
- Integrated Mapbox Geocoding API and Cloudinary CDN for location-based listing discovery and optimized cloud media storage, improving user experience and asset delivery.
- Developed complete CRUD functionality for 50+ travel listings with server-side validation, centralized error handling, flash messaging, reviews, and booking workflows following MVC architecture.

TECHNICAL SKILLS

Programming Languages: C++, JavaScript (ES6+), TypeScript, Python

Frontend Technologies: React.js, Next.js, HTML5, CSS3, Tailwind CSS, EJS

Backend Technologies: Node.js, Express.js, RESTful API Development, JWT Authentication, Role-Based Access Control (RBAC), Middleware Architecture, Microservices

Databases: MongoDB, PostgreSQL, Mongoose ODM, Database Schema Design, Query Optimization, Database Indexing

Cloud & DevOps: Docker, Git, GitHub, CI/CD, Linux, Vercel, AWS EC2, AWS S3, Postman

Artificial Intelligence & APIs: OpenAI GPT-4o API, Google Cloud Vision API, OCR Integration, Prompt Engineering

Developer Tools: Visual Studio Code, GitHub Copilot, npm, Cloudinary, Mapbox API, Multer

Software Engineering Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design, Scalable System Architecture, Secure Coding, MVC Architecture, Version Control, Agile/Scrum

ACHIEVEMENTS & RECOGNITION

•Software Engineering Intern– Noventiq (Umbrella Infotech), Noida

Dec 2024 – Feb 2025

- Competitively selected for a Software Engineering internship; contributed to enterprise cloud-native backend systems, RESTful API development, Docker-based deployments, secure authentication, and Agile software delivery.

•3 Production-Grade Full-Stack Applications– AI, Healthcare, Travel & B2B Platforms

2023 – 2025

- Architected and delivered three end-to-end software products featuring 20+ RESTful APIs, AI-powered OCR, GPT-4o integration, JWT authentication, RBAC authorization, Docker containerization, and scalable backend architecture.

•Open Source Portfolio– GitHub: github.com/Vardxn

2023 – Present

- Maintained 5+ public repositories demonstrating full-stack software engineering, backend architecture, REST API development, secure authentication, database design, and production-ready deployment practices.

•Qualified JEE Main 2022– National-Level Engineering Entrance Examination

2022

- Qualified JEE Main 2022, securing admission to the B.Tech Information Technology program at the National Institute of Technology (NIT) Srinagar through India's national engineering entrance examination.

CERTIFICATIONS

•Full Stack Web Development – Sigma 8.0– Apna College by Aman Dhatarwal (Completed)

2024

- Completed an industry-oriented Full Stack Software Engineering curriculum covering Data Structures & Algorithms, MERN Stack, RESTful API Development, Authentication & Authorization, System Design, Database Design, and end-to-end application deployment.

•AI/ML Prime– Apna College by Aman Dhatarwal (In Progress)

2024 – Present

- Learning Machine Learning fundamentals, Python for Data Science, NumPy, Pandas, Matplotlib, data preprocessing, model development, and AI-powered application development.

•Software Engineering Job Simulation– JPMorgan Chase & Co. via Forage (Completed)

Jan 2026

- Completed enterprise software engineering simulations involving Kafka integration, H2 in-memory databases, RESTful API development, backend controller implementation, debugging, and production-oriented software engineering workflows.